Math 343 / 643 Spring 2026
Final Examination

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May 20, 2026

Full Name ____

Code of Academic Integrity

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Activities that have the effect or intention of interfering with education, pursuit of knowledge, or fair evaluation of a student's performance are prohibited. Examples of such activities include but are not limited to the following definitions:

Cheating Using or attempting to use unauthorized assistance, material, or study aids in examinations or other academic work or preventing, or attempting to prevent, another from using authorized assistance, material, or study aids. Example: using an unauthorized cheat sheet in a quiz or exam, altering a graded exam and resubmitting it for a better grade, etc.

I acknowledge and agree to uphold this Code of Academic Integrity.

______________________________________ ________________

signature date

Instructions

This exam is 120 minutes (variable time per question) and closed-book. You are allowed **three** pages (front and back) of "cheat sheets", blank scrap paper (provided by the proctor) and a graphing calculator (which is not your smartphone). Please read the questions carefully. Within each problem, I recommend considering the questions that are easy first and then circling back to evaluate the harder ones. No food is allowed, only drinks.

Problem 1 You are trying to model daily average calorie proportion in people's diet from ultra high processed foods (UHPFs). Since it's a proportion, the response is $\mathcal{Y} = [0, 1]$. Typically, such responses are modeled with the beta rv. We will reparameterize it below from what you saw in MATH 340:

$$Y \sim \text{Beta}\,(\mu,\, \phi) := \frac{1}{B(\mu\phi, (1-\mu)\phi)}\, y^{\mu\phi - 1}(1-y)^{(1-\mu)\phi - 1}\mathbb{1}_{y>0}, \quad \mathbb{E}\,[Y] = \mu$$

This way we can model its expectation directly. The ϕ parameter is then a nuisance parameter controlling the variance (the thinness of the PDF's bell shape).

There is another wrinkle: some people are very healthy and eat no UHPFs; some people are extremely unhealthy and eat all UHPFs and everything in between. Thus the realized values of the y_i's in the survey will have 0, 1's and any fractions between 0 and 1. Thus we need to model exact 0's and exact 1's. We can do that with two more parameters:

$$Y_i \overset{ind}{\sim} \begin{cases} 0 & w.p. \quad \theta_0 \\ \text{Beta}\,(\mu,\, \phi) & w.p. \quad 1 - \theta_0 - \theta_1 \\ 1 & w.p. \quad \theta_1 \end{cases}$$

This model is known as the *zero-one inflated beta model*. Also, we have covariates that we care about modeling the effect for each individual i:

- $x_{i,1}$: A continuous metric called the Household Income-to-Poverty Ratio
- $x_{i,2}$: A continuous metric called the Food Desert Index / Neighborhood Food Environment which is geographic proximity to supermarkets selling healthy food
- $x_{i,3}$: A binary variable assessing whether or not WIC/SNAP enrolled
- $x_{i,4}$: Average weekly working hours scaled over 40
- $x_{i,5}$: Continuous metric of age in years scaled over 120
- $x_{i,6}$: Continuous metric of body-to-mass-index scaled over 30
- $x_{i,7}$: Binary variable assessing the presence of a comorbidity such as diabetes, high blood pressure, etc.

We would like to model the mean of the non-zero, non-one proportion with these features. But we do not model the proportion of zeroes and ones with these features. So we choose a linear model and we use a logit link function just like in logistic regression:

$$\mu_i = \phi(\boldsymbol{x}_i\boldsymbol{\beta}) = \frac{e^{\beta_0 + \beta_1 x_{i,1} + \ldots + \beta_7 x_{i,7}}}{1 + e^{\beta_0 + \beta_1 x_{i,1} + \ldots + \beta_7 x_{i,7}}} = \frac{1}{1 + e^{-\boldsymbol{x}_i\boldsymbol{\beta}}}, \quad 1 - \mu_i = \frac{1}{1 + e^{\boldsymbol{x}_i\boldsymbol{\beta}}}$$

Also, for convenience, let

$$n_0 := \sum_{i=1}^{n} \mathbb{1}_{y_i = 0} \quad \text{and} \quad n_1 := \sum_{i=1}^{n} \mathbb{1}_{y_i = 1} \quad \text{and} \quad n_p := \sum_{i=1}^{n} \mathbb{1}_{y_i \in (0,1)} \;\; \text{where} \;\; n = n_0 + n_1 + n_p.$$

We have n observations $< \boldsymbol{x}_1, y_1 >, \ldots, < \boldsymbol{x}_n, y_n >$ pairs stacked like in 342 as matrix $\boldsymbol{X}$, vector $\boldsymbol{y}$ where each subject i has $p = 7$ measurements in row vector $\boldsymbol{x}_i \in \mathbb{R}^p$ and $y_i \in [0, 1]$.

(a) [2 pt / 2 pts] How many scalar parameters can we draw inference for in this model?

$p+1$ for the β_j's, one for the θ_0, one for θ_1 and one for ϕ yields $p+4 = 11$ total features.

(b) [5 pt / 7 pts] Write the likelihood function explicitly from the definition of the zero-one inflated beta model with the logit glm link. Simplify as much as possible.

$$
\begin{aligned}
\mathcal{L}(\boldsymbol{\beta}, \theta_0, \theta_1, \phi; \boldsymbol{X}, \boldsymbol{y}) &= \prod_{i=1}^{n} \theta_0^{\mathbb{1}_{y_i=0}} \theta_1^{\mathbb{1}_{y_i=1}} \times \\
&\quad \left((1-\theta_0-\theta_1) \frac{1}{B(\mu_i\phi, (1-\mu_i)\phi)}\, y^{\mu_i\phi-1}(1-y)^{(1-\mu_i)\phi-1} \right)^{\mathbb{1}_{y_i\in(0,1)}} \\
&= \theta_0^{n_0}\theta_1^{n_1}(1-\theta_0-\theta_1)^{n_p} \times \\
&\quad \prod_{i=1}^{n} \left(\frac{1}{B(\mu_i\phi, (1-\mu_i)\phi)}\, y^{\mu_i\phi-1}(1-y)^{(1-\mu_i)\phi-1} \right)^{\mathbb{1}_{y_i\in(0,1)}} \\
&= \theta_0^{n_0}\theta_1^{n_1}(1-\theta_0-\theta_1)^{n_p} \times \\
&\quad \prod_{i=1}^{n} \left(\frac{1}{B\left(\frac{\phi}{1+e^{-\boldsymbol{\beta}\boldsymbol{x}_i}}, \frac{\phi}{1+\phi e^{\boldsymbol{\beta}\boldsymbol{x}_i}}\right)}\, y^{\frac{\phi}{1+e^{-\boldsymbol{\beta}\boldsymbol{x}_i}}-1}(1-y)^{\frac{\phi}{1+e^{\boldsymbol{\beta}\boldsymbol{x}_i}}-1} \right)^{\mathbb{1}_{y_i\in(0,1)}}
\end{aligned}
$$

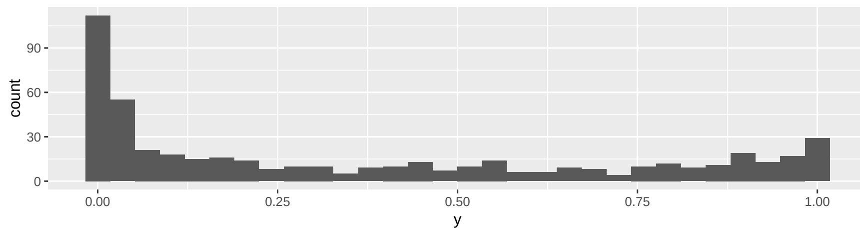
Assume for the rest of the problem that the prior is Laplace i.e. $f(\boldsymbol{\beta}, \theta_0, \theta_1, \phi) \propto 1$.

(c) [2 pt / 9 pts] Are all the conditional distributions for the parameters proportional to known distributions? Yes / No

(d) [2 pt / 11 pts] What are the names of the two methods we studied in class to allow for computational Bayesian inference in this scenario?

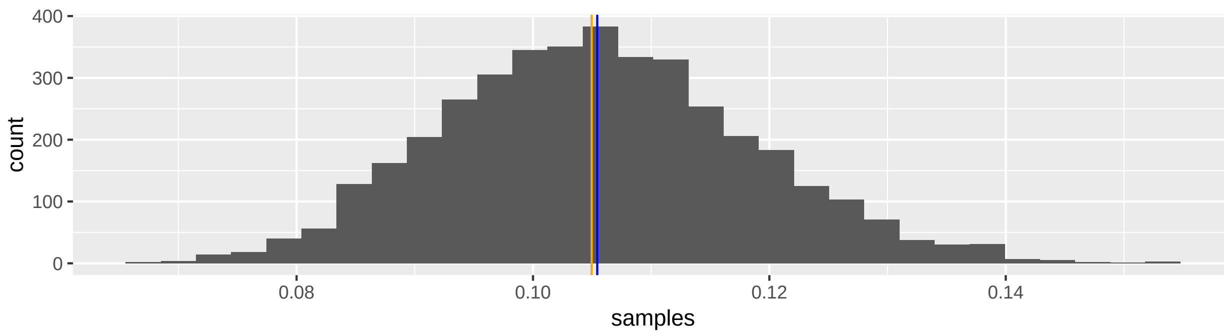
Metropolis-Hastings sampling, [No U-Turn Sampling within] Hamiltonian MCMC

We now turn to the samples and the features. We have $n = 500$ samples of surveys. Here is a histogram of the raw data.



We now use `stan` to do the computational inference. We draw 5,000 samples in `stan`.

Below is a histogram of the samples for θ_0 produced by our visualize chains functions we used in class:



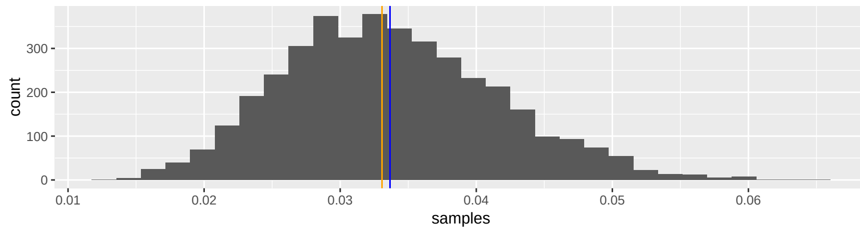
(e) [2 pt / 13 pts] Approximate θ_0^{MMAE}.

0.105

(f) [3 pt / 16 pts] Approximate $CR_{\theta_0,95\%}$.

$CR_{\theta_0,95\%} = [0.082, 0.132]$

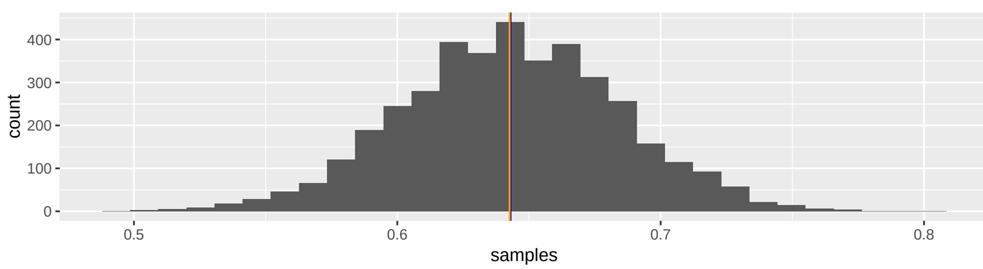
Below is a histogram of the samples for θ_1 produced by our visualize chains functions we used in class:

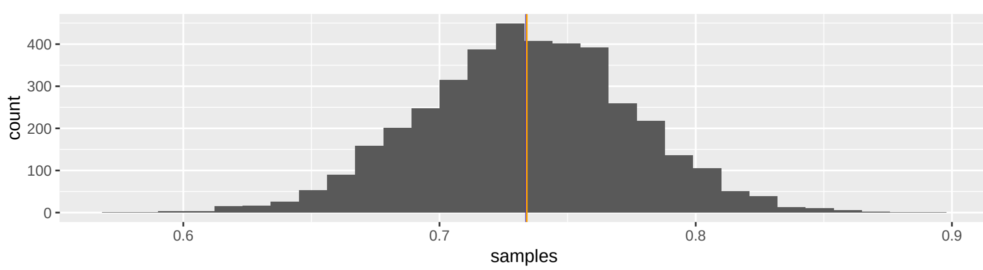


(g) [4 pt / 20 pts] Approximate a decision for $H_0 : \theta_0 = \theta_1$ at $\alpha = 5\%$. Justify your decision.

$\boxed{\text{Reject } H_0}$. To do this test properly, we would need to tally $\theta_0 - \theta_1$ samples across the MCMC, define a δ equivalence and integrate the proportion of samples in $[0 \pm \delta]$. Since none of these samples in both picture are anywhere near zero, we reject for any reasonable δ.

Below is a histogram of the samples for β_1 and β_7 produced by our visualize chains functions we used in class:





(h) [5 pt / 25 pts] Make a decision about the omnibus test (i.e., the null hypothesis is that none of the $p = 7$ features matter in the linear model). Justify your decision.

Since (a) $H_0 : \beta_1 = 0$, $H_0 : \beta_7 = 0$ will be rejected at any reasonable δ or (b) CR's for β_1, β_7 will not include zero at any reasonable α, we can assume that we must also $\boxed{\text{reject}}$ the omnibus null hypothesis that $H_0 : \beta_1 = \ldots = \beta_7 = 0$.

Here are the averages and standard deviations of the beta samples from `stan`:

```
        avg   sd
beta_0 -0.616 0.039
beta_1  0.643 0.042
beta_2  0.667 0.040
beta_3  0.756 0.042
beta_4  0.705 0.043
beta_5  0.657 0.042
beta_6  0.741 0.041
beta_7  0.734 0.041
```

(i) [7 pt / 32 pts] For a new person with $\boldsymbol{x}_\star = [1.3, 0.2, 1, 0.6, 0.4, 0.1, 0]$, estimate the proportion of calories they consume from UHPFs. Round to nearest 2 digits.

$$
\begin{aligned}
\mathbb{E}[Y \mid \boldsymbol{x}_\star] &= \hat{\hat{\theta}}_0(0) + \hat{\hat{\theta}}_1(1) + (1 - \hat{\hat{\theta}}_0 - \hat{\hat{\theta}}_1)\, \mathbb{E}[Y \mid \boldsymbol{x}_\star, Y \in (0,1)] \\
&= 0.105(0) + 0.033(1) + (1 - 0.105 - 0.033)\, \mathbb{E}[Y \mid \boldsymbol{x}_\star, Y \in (0,1)] \\
&= 0.033 + 0.862\, \hat{\hat{\mu}}(\boldsymbol{x}_\star) \\
&= 0.033 + 0.862\, \frac{1}{1 + e^{-\boldsymbol{\beta}\boldsymbol{x}_\star}} \\
&= 0.033 + 0.862\, \frac{1}{1 + e^{-[-0.616, 0.643, 0.667, 0.756, 0.705, 0.657, 0.741, 0.734]^\top [1, 1.3, 0.2, 1, 0.6, 0.4, 0.1, 0]}} \\
&= 0.033 + 0.862\, \frac{1}{1 + e^{-1.8692}} \\
&= 0.033 + 0.862\, \frac{1}{1.154247} \\
&= 0.78
\end{aligned}
$$

(j) [6 pt / 38 pts] Assuming the data was collected using a simple random sample of subjects, provide an interpretation of the estimated coefficient of x_3, the binary variable assessing whether or not the subject is `WIC/SNAP` enrolled. Remember, the glm scale is "logit proportion of calories from UHPFs".

When comparing two subjects (A) and (B) which are sampled in the same fashion as the other subjects in this dataset where (A) has `WIC/SNAP` and (B) does not have `WIC/SNAP` but share the same other observed measurements otherwise, then (A) is predicted to have an estimated logit proportion of calories from UHPFs 0.756 ± 0.042 higher than (B)'s logit proportion of calories from UHPFs assuming the log odds of calories from UHPFs is linear in the measurements considered herein and remains stationary.

Instead of the Laplace prior $f(\boldsymbol{\beta}, \theta_0, \theta_1, \phi) \propto 1$, we now consider the following prior:

$$
\begin{aligned}
\theta_0 &\sim \text{Beta}\left(\frac{1}{2}, \frac{1}{2}\right), \quad \theta_1 \sim \text{Beta}\left(\frac{1}{2}, \frac{1}{2}\right) \\
\phi &\propto 1/\phi^2 \\
\beta_0 &\propto 1 \\
\beta_1 &\sim \text{Laplace}(0,1), \quad \beta_2 \sim \text{Laplace}(0,1), \ldots, \quad \beta_7 \sim \text{Laplace}(0,1)
\end{aligned}
$$

Then assume these priors and resample the posterior from `stan`.

(k) [6 pt / 44 pts] Compared to the estimates for $\beta_1, \ldots, \beta_7$ from the "averages and standard deviations of the beta samples from `stan`" table previously shown to you, what changes you expect to see for the new estimates for $\beta_1, \ldots, \beta_7$ with these new priors?

The Laplace prior is equivalent to an L1 penalty (AKA the LASSO regression) hence we would expect some of the β_j estimates will be zero and the estimates that are nonzero will be closer to zero. Although we studied this for the linear model for continuous responses, this behavior transfers over to glm's as well.

(l) [7 pt / 51 pts] If we were to retain the zero-one inflated beta model, but instead use Frequentist inference, explain in detail (step-by-step) how you would approximate $CI_{\beta_7,95\%}$. Do not do any actual numeric computations. Write in clear, neat English with only the bare mathematical notation you need to explain your idea properly. Hint: assume your answer from part (b) is correct.

You would take the log of your answer from (b) to obtain the log likelihood. Then you would find its gradient. Then you would use a numerical optimization to find the MLE for all parameters by finding the roots of the gradient. Then pull out the last entry of the MLE estimate which we will denote $\hat{\hat{\beta}}_7^{MLE}$. Then you would find the Hessian from the gradient. The negative Hessian is then approximately the Fisher Information matrix. Then you invert the Fisher Information matrix estimate and pull out the diagonal entry of that inverted matrix corresponding to the β_7 component, i.e., the 14,14 element, denote that quantity $s^2_{\beta_7}$. Then you would compute:

$$\hat{\hat{CI}}_{\beta_7,95\%} \approx \left[\hat{\hat{\beta}}_7^{MLE} \pm 1.96 s_{\beta_7}\right]$$

Alternatively, you can use the bootstrap.

(m) [7 pt / 58 pts] We now wish to create a model that also allows these $p = 7$ features to influence θ_0, θ_1. Use a glm. Write out its likelihood below. There are many correct answers. Hint: use the first line from your answer to (b).

Here is the answer from the first line of (b):

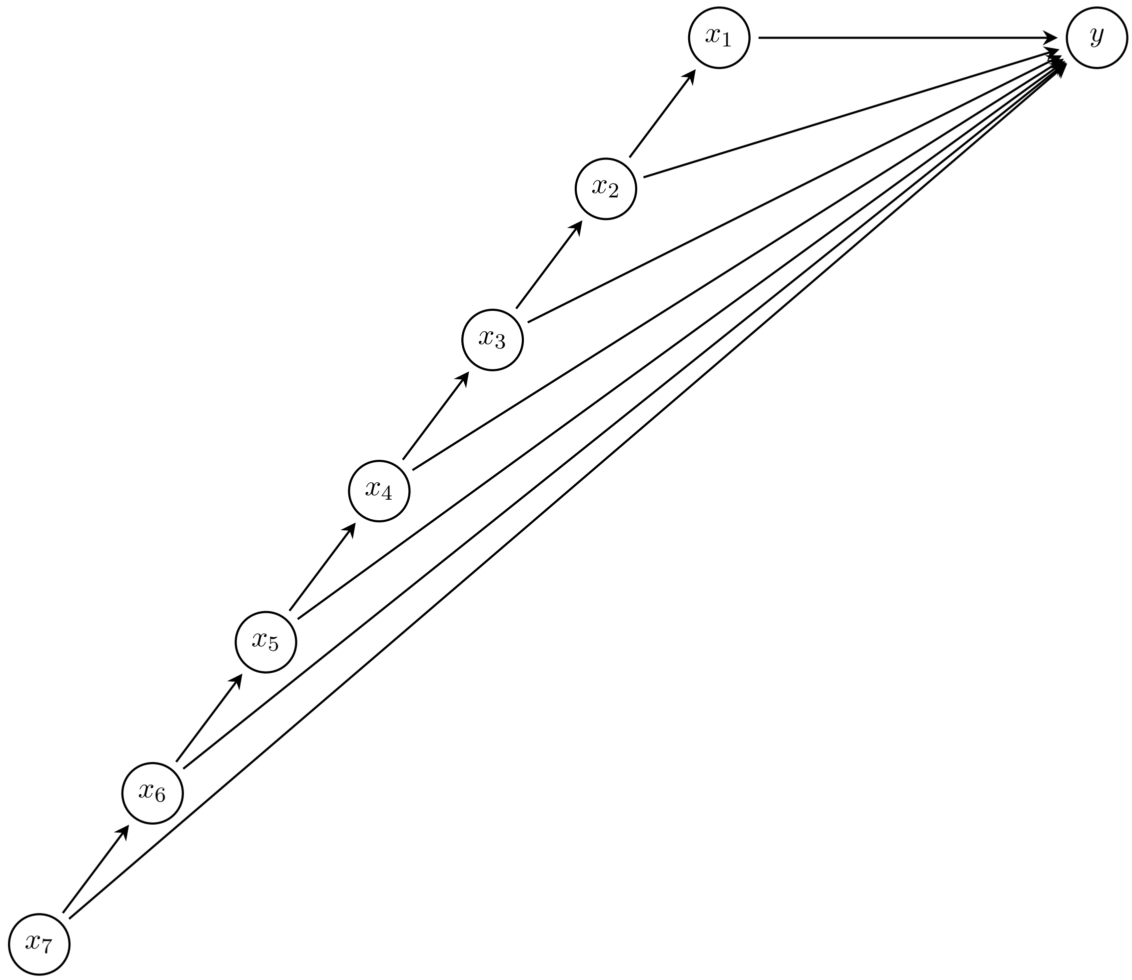
$$\begin{aligned}\mathcal{L}(\boldsymbol{\beta}, \theta_0, \theta_1, \phi; \boldsymbol{X}, \boldsymbol{y}) &= \prod_{i=1}^{n} \theta_0^{\mathbb{1}_{y_i=0}} \theta_1^{\mathbb{1}_{y_i=1}} \times \\ &\left((1-\theta_0-\theta_1)\frac{1}{B(\mu_i\phi, (1-\mu_i)\phi)} y^{\mu_i\phi-1}(1-y)^{(1-\mu_i)\phi-1}\right)^{\mathbb{1}_{y_i\in(0,1)}}\end{aligned}$$

We now want to allow the features to specify θ_0, θ_1 via a glm. Since these parameters are probabilities, we again can use the logit link. Let $\boldsymbol{\gamma}$ be the $p+1$ linear parameters for θ_0 and let $\boldsymbol{\eta}$ be the $p+1$ linear parameters for θ_1, substituting the formula from page 1 for both θ_0, θ_1, we also note that we cannot use the simplification with n_0, n_1 anymore:

$$\begin{aligned}\mathcal{L}(\boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\eta}, \phi; \boldsymbol{X}, \boldsymbol{y}) &= \prod_{i=1}^{n} \left(\frac{1}{1+e^{-\boldsymbol{\gamma}\boldsymbol{x}_i}}\right)^{\mathbb{1}_{y_i=0}} \left(\frac{1}{1+e^{-\boldsymbol{\eta}\boldsymbol{x}_i}}\right)^{\mathbb{1}_{y_i=1}} \times \\ &\left(\left(1-\frac{1}{1+e^{-\boldsymbol{\gamma}\boldsymbol{x}_i}}-\frac{1}{1+e^{-\boldsymbol{\eta}\boldsymbol{x}_i}}\right) \times \right. \\ &\left.\frac{1}{B\left(\frac{\phi}{1+e^{-\boldsymbol{\beta}\boldsymbol{x}_i}}, \frac{\phi}{1+\phi e^{\boldsymbol{\beta}\boldsymbol{x}_i}}\right)} y^{\frac{\phi}{1+e^{-\boldsymbol{\beta}\boldsymbol{x}_i}}-1}(1-y)^{\frac{\phi}{1+e^{\boldsymbol{\beta}\boldsymbol{x}_i}}-1}\right)^{\mathbb{1}_{y_i\in(0,1)}}\end{aligned}$$

Regardless of what you wrote above, we will not be using this model for the rest of the exam. Assume the original model where θ_0, θ_1 are not estimated using covariates.

Assume the following DAG for our DGP:



(n) [3 pt / 61 pts] Which variable(s) are causal for the response?

$x_1, x_2, \ldots, x_7$

(o) [4 pt / 65 pts] Why would a zero-one beta regression of $y \sim x_1$ provide a biased causal estimate for x_1's effect on y?

$\boxed{\text{No}}$. Because x_2 is a common cause of both x_1, y.

(p) [3 pt / 68 pts] Which paradox may occur in the zero-one beta regression of $y \sim x_1$? You just need to write the name of the paradox below (one word).

Simpson's paradox

(q) [3 pt / 71 pts] Assume you run two regressions [A] $y \sim x_1 + x_2$ and [B] $y \sim x_1$. Is it possible for b_1 to be different signs in [A] vs [B]? Yes / No

(r) [4 pt / 75 pts] Why would a zero-one beta regression of $y \sim x_1 + x_2$ provide a biased causal estimate for x_2's effect on y?

Because x_1 partially blocks the effect of x_2 on y.

(s) [5 pt / 80 pts] Which regression can provide an unbiased causal estimate of x_1 on y? Use the R formula notation beginning with "$y \sim \ldots$" using "+" between covariates.

$y \sim x_1 + x_2 + \ldots + x_7$

Assume we no longer believe in this DAG. And assume we are now interested in the causal variable x_3, the binary variable assessing whether or not the subject is `WIC/SNAP` enrolled.

(t) [3 pt / 83 pts] Is it possible to run a regression (any regression) on the data you have and be sure you can get a causal estimate for x_3? Yes / No

Regardless of what you wrote in the previous question, we are considering doing a randomized two-arm controlled trial (RCT) on the variable $w = x_3$.

(u) [4 pt / 87 pts] Are there any practical or ethical concerns with this RCT?

Yes, you cannot ethically give people WIC/SNAP because it would waste taxpayer money.

Regardless of what you wrote in the previous question, assume the data in this exam came from a randomized two-arm controlled trial (RCT) on the variable $w = x_3$.

(v) [7 pt / 94 pts] Provide an interpretation of the estimated coefficient of $w = x_3$, the binary variable assessing whether or not the subject is `WIC/SNAP` enrolled. Remember, the glm scale is "logit proportion of calories from UHPFs".

If a person receives `WIC/SNAP` and all other characteristics about this person stays constant, then the estimated logit proportion of calories from UHPFs will resultingly increase in 0.756 ± 0.042 assuming the logit proportion of calories from UHPFs is linear in the measurements considered herein and remains stationary.

(w) [3 pt / 97 pts] Treating the y as continuous metric, assuming homoskedastic error and an additive treatment effect and ignoring the covariates, what is the optimal design?

BCRD: 250 subjects in each arm

(x) [3 pt / 100 pts] In this case, would the estimator $\bar{Y}_T - \bar{Y}_C$ (the difference of the average proportion of calories form UHPFs in the `WIC/SNAP` $= 1$ group and the average proportion of calories form UHPFs `WIC/SNAP` $= 0$ group) be unbiased for the population additive average treatment effect? Yes / No